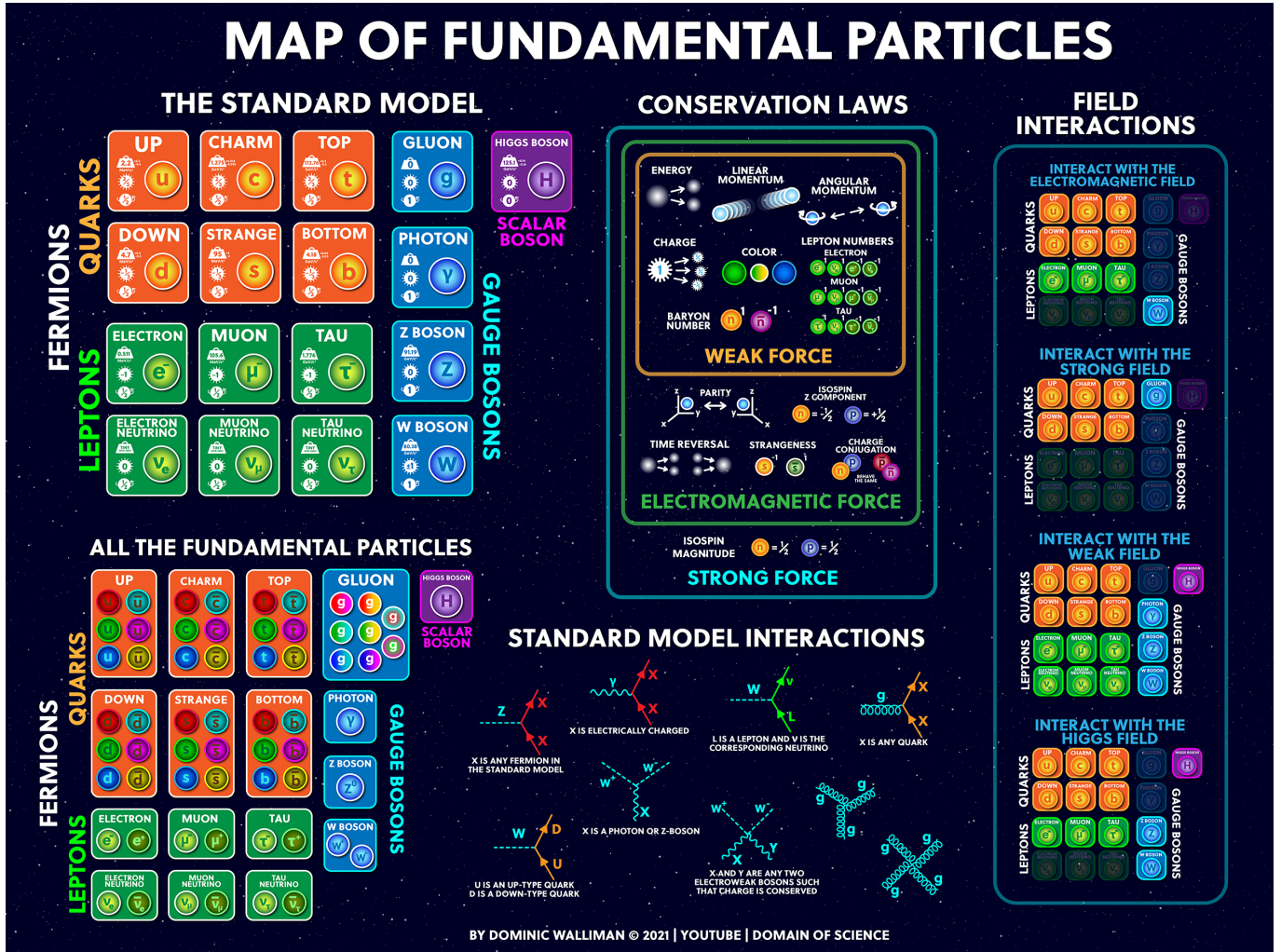


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1 Fundamental Particles



1.1 Units

To simplify calculations we switch to natural units, where $c = \hbar = 1$, $k = 1/4\pi$, and therefore all units are in powers of eV. We can then say that

$$(NU) = \hbar^a c^b (SI)$$

and with that go between systems. Some good conversions to remember is:

$$\hbar c \approx 197 \text{ MeV} \cdot \text{fm}, \quad (1.17)$$

$$\hbar \approx 6.58 \cdot 10^{-16} \text{ eV} \cdot \text{s}. \quad (1.18)$$

2 Scattering and Decay

2.1 Special Relativity

Covariance is a prescription on the form of the physical laws that ensures compliance with the axioms of relativity:

(A1) The physical laws retain the same functional form in all inertial frames.

(A2) The speed of light in vacuum is the same for all inertial observers.

The Four-vector for location is

$$x^\mu = (ct, x, y, z), \quad (2.19)$$

and the transformed vector is:

$$\begin{pmatrix} x'_0 \\ x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad (2.21)$$

$$x'^\mu = \Lambda^\mu_\nu x^\nu. \quad (2.22)$$

The velocity and acceleration four vectors are:

$$\begin{aligned} v^\mu &= \gamma(c, v_x, v_y, v_z), \\ a^\mu &= \gamma(\dot{\gamma}c, \dot{\gamma}\mathbf{v} + \gamma\mathbf{a}). \end{aligned}$$

The four-momentum is:

$$p^\mu = (m\gamma, m\gamma\mathbf{v}) = (E, \mathbf{p}), \quad (2.40)$$

We can use this to calculate energies in particle interactions, since we know there is conservation of momentum.

$$\sum_{i=1}^{N_i} p_i^\mu = \sum_{j=1}^{N_f} p_j^\mu. \quad (2.43)$$

Some helpful equations are:

$$(p^\mu)^2 = (p^0)^2 - |\mathbf{p}|^2, \quad (2.57)$$

$$E = p^0 = (m^2 + |\mathbf{p}|^2)^{1/2}. \quad (2.58)$$

2.2 Two-body decay

Creation or destruction of particles can only happen if kinematics of Special relativity and of laws of the standard model are conserved. For decays of one particle a into two particles, the momentum must be conserved:

$$p_a^\mu = p_1^\mu + p_2^\mu, \quad (2.64)$$

if we look at the rest frame of a , we get that

$$p_2^2 = (p_a - p_1)^2 = p_a^2 + p_1^2 - 2p_a \cdot p_1, \quad (2.65)$$

$$m_2^2 = m_a^2 + m_1^2 - 2p_a^\mu \cdot p_{1\mu}. \quad (2.66)$$

Here we used that the norm of a single particles momentum is it's rest mass. Because we are in the rest frame of a

$$p_a^\mu p_{1\mu} = g_{\mu\nu} p_a^\mu p_1^\nu = m_a E_1, \quad (2.67)$$

and with this we have:

$$E_1 = \frac{m_a^2 + m_1^2 - m_2^2}{2m_a}, \quad E_2 = \frac{m_a^2 + m_2^2 - m_1^2}{2m_a}. \quad (2.68)$$

Three-body decay

In a three-body decay $a \rightarrow 1 + 2 + 3$, we have too many variables, and can therefore only say something about the minimum and maximum energy of particle 1.

$$m_1 \leq E_1^{CM} \leq \frac{m_a^2 + m_1^2 - (m_2 + m_3)^2}{2m_a}. \quad (2.71)$$

2.3 Decay Amplitudes

In a general term we can use the **relativistic golden rule** for unstable particles that decay into N_f particles. We use this rule to calculate the transition rate, which is the probability per unit time that the particle in the initial state decay into the final state:

$$\Gamma_{i \rightarrow f} = \frac{1}{2E_a} \int \prod_{i=1}^{N_f} \frac{d^4 p_i}{(2\pi)^4} |\mathcal{M}|^2 (2\pi)^4 \delta^4 \left(p_a - \sum_{j=1}^{N_f} p_j \right) \times (2\pi) \delta(p_i^2 - m_i^2). \quad (2.90)$$

The decay Lorentz transforms as

$$\Gamma'_{i \rightarrow f} = \Gamma_{i \rightarrow f} / \gamma. \quad (2.94)$$

And is related to the lifetime of a particle by

$$\tau = 1/\Gamma. \quad (2.96)$$

But most particles decay in multiple nodes and we therefore have to generalize as:

$$\tau = \frac{1}{\Gamma} = \frac{1}{\sum_{i=1}^N \Gamma_i}. \quad (2.101)$$

The **Branching Ratio** of node i is defined as:

$$\text{BR}(i) = \frac{\Gamma_i}{\Gamma}. \quad (2.102)$$

In (2.90) we have $|\mathcal{M}|^2$ which is the Lorentz-invariant matrix element for n-body decay:

$$\mathcal{M} = (2E_a)^{1/2} \prod_{i=1}^{N_f} (2E_i)^{1/2} \mathcal{T}, \quad (2.79)$$

where $\mathcal{T} = \langle f | \hat{H} | i \rangle$

2.4 Scattering

Most of the formulas for decay holds for two-body scattering $a + b \rightarrow 1 + \dots + n$, we just have to make a few corrections. A measurement for scattering in QM is the **differential cross section**, which is the ratio between the number of particles that are scattered in a solid angle between Ω and $\Omega + d\Omega$ per unit time divided by the flux (the rate of incoming particles per unit surface). Or in other words the likelihood that a particle is scattered at an angle Ω .

$$\frac{d\sigma}{d\Omega}(E, \Omega) = \frac{1}{F} \frac{dN_s}{d\Omega}. \quad (2.113)$$

This leads to the **relativistic golden rule for the cross section**

$$\begin{aligned} \sigma_{i \rightarrow f} &= \frac{(2\pi)^4}{2E_a 2E_b |v_a - v_b|} \int \prod_{i=1}^n \frac{d^3 \mathbf{p}_i}{(2\pi)^3 2E_i} \\ &\times |\mathcal{M}|^2 \delta \left(E_a + E_b - \sum_{j=1}^n E_j \right) \delta^3 \left(\mathbf{p}_a + \mathbf{p}_b - \sum_{j=1}^n \mathbf{p}_j \right), \end{aligned} \quad (2.128)$$

but here we have a new $\mathcal{M}$

$$\mathcal{M} = (2E_a)^{1/2} (2E_b)^{1/2} \prod_{i=1}^n (2E_i)^{1/2} \mathcal{T}. \quad (2.127)$$

A usefull Lorentz-invariant is the **first Mandelstam variable** for two-body scattering

$$s = (p_a + p_b)^2 = (p_{tot})_\mu (p_{tot})^\mu, \quad (2.131)$$

where $(p_{tot})^\mu = p_a^\mu + p_b^\mu$ is the total initial momentum. $\sqrt{s}$ is often called invariant mass. The goal of QFT is to calculate $\mathcal{M}$ and in most cases $|\mathcal{M}|^2$ is a smooth and slowly variable function of s . If $|\mathcal{M}|^2 \approx \text{constant}$ then

$$\sigma = \frac{1}{s}, \quad (2.136)$$

for $\sqrt{s} \gg m_a, m_b$.

3 Symmetries

3.1 Symmetries in Quantum Mechanics

In quantum mechanics (QM), a symmetry is a set of transformations represented by **unitary** (or anti-unitary) **operators**, U , that leave physical observables

unchanged. This requires $U^\dagger U = I$. A system's Hamiltonian, H , is invariant under a symmetry transformation if it commutes with the symmetry operator, $[H, U] = 0$. **Wigner's theorem** states that all symmetries in QM are represented by such unitary or anti-unitary operators that commute with the Hamiltonian. If $[H, U] = 0$, the expectation value $\langle U \rangle$ is constant in time, making it an integral of motion, though not necessarily a directly measurable conserved physical quantity since U may not be Hermitian.

3.2 Classification of Symmetries

Symmetries in QM can be **classified** into four types:

- **Continuous external:** Depend smoothly on parameters and change space-time coordinates (e.g., translations, rotations).
- **Continuous internal:** Depend smoothly on parameters but *do not* change space-time coordinates.
- **Discrete external:** Involve a finite set of transformations that change space-time coordinates (e.g., parity P).
- **Discrete internal:** Involve a finite set of transformations that *do not* change space-time coordinates (e.g., charge conjugation C).

Continuous symmetries described by unitary operators $U(a_j)$ can be expressed exponentially in terms of **Hermitian generators**

$$T_j : U(a_1 \dots a_N) = e^{i \sum_j a_j T_j}. \quad (5.9)$$

Because these generators T_j are Hermitian and commute with the Hamiltonian H , their expectation values $\langle T_j \rangle$ represent **conserved physical quantities**.

3.3 Inversion

Inversions are discrete transformations, the unitary operator has to fulfill

$$U^2 = I, \quad (5.26)$$

and if the operator is applied twice we should return to the original system. One such symmetry could be **parity** where we invert $\mathbf{x}$ turning a right hand system to a left hand system.

$$\mathcal{P}|x\rangle = |-x\rangle, \quad (5.27)$$

which in quantum mechanics is a unitary, hermitian operator

$$\mathcal{P}\psi(\mathbf{x}) = \psi(-\mathbf{x}). \quad (5.28)$$

The parity of a system is only defined if the wavefunction is an eigenstate of $\mathcal{P}$, with eigenvalues $+1$ (even parity) or -1 (odd parity).

3.4 The Intrinsic Parity of Particles

In QM we define the **intrinsic parity** of a particle, as the eigenvalue of the wavefunction of the particle in its restframe. For a photon we can not find the intrinsic parity directly with QM, so we look at the decay of excited hydrogen:

$$^1_1\text{H}^* \rightarrow ^1_1\text{H} + \gamma, \quad (5.34)$$

We know that the parity of hydrogen is

$$\mathcal{P}\psi_{^1_1\text{H}} = (-1)^l \psi_{^1_1\text{H}}$$

4 Appendix

4.1 Four-Vector

A four-vector a^μ is an object in Minkowski space-time with four components. It is written as:

$$a^\mu = (a^0, a^1, a^2, a^3) = (a^0, \mathbf{a})$$

where a^0 is the time-like component and $\mathbf{a} = (a^1, a^2, a^3)$ is the vector of the space-like components.

Contravariant and Covariant Vectors

A four-vector can be in two forms:

- **Contravariant** (upper index): $a^\mu = (a^0, a^1, a^2, a^3)$
- **Covariant** (lower index): $a_\mu = (a_0, a_1, a_2, a_3)$

The transformation between them uses the Minkowski metric tensor $g_{\mu\nu}$.

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (2.31)$$

The rule to lower an index is $a_\mu = g_{\mu\nu}a^\nu$. This results in:

$$a_\mu = (a^0, -a^1, -a^2, -a^3) = (a^0, -\mathbf{a})$$

Scalar Product

The scalar product of two four-vectors a^μ and b^μ is a Lorentz invariant scalar, denoted as $a \cdot b$ or $a_\mu b^\mu$.

$$a \cdot b = a_\mu b^\mu = g_{\mu\nu}a^\mu b^\nu \quad (1)$$

Using the Einstein summation convention, this expands to:

$$a \cdot b = a^0 b^0 - a^1 b^1 - a^2 b^2 - a^3 b^3 = a^0 b^0 - \mathbf{a} \cdot \mathbf{b}$$

Squared Norm of a Four-Vector

The squared norm of a four-vector a^μ is its scalar product with itself, denoted a^2 .

$$a^2 = a_\mu a^\mu = (a^0)^2 - (a^1)^2 - (a^2)^2 - (a^3)^2 = (a^0)^2 - |\mathbf{a}|^2 \quad (2)$$

The sign of the squared norm classifies the four-vector:

- $a^2 > 0$: **Time-like**
- $a^2 < 0$: **Space-like**
- $a^2 = 0$: **Light-like** or **null**

The Four-Momentum

The four-momentum p^μ is a key four-vector in relativistic kinematics. In natural units ($c = 1$), it is defined as:

$$p^\mu = (E, \mathbf{p}) = (m\gamma, m\gamma\mathbf{v}) \quad (2.40)$$

where $E = m\gamma$ is the relativistic energy, $\mathbf{p} = m\gamma\mathbf{v}$ is the relativistic three-momentum, m is the rest mass, and $\gamma = 1/\sqrt{1-v^2}$ is the Lorentz factor.

The squared norm of the four-momentum is the squared rest mass, an invariant:

$$p^2 = p_\mu p^\mu = E^2 - |\mathbf{p}|^2 = m^2$$

This yields the energy-momentum relation:

$$E = \sqrt{|\mathbf{p}|^2 + m^2} \quad (2.58)$$

Lorentz Transformations

A four-vector a^μ transforms between inertial frames via a Lorentz transformation matrix Λ_ν^μ .

$$a'^\mu = \Lambda_\nu^\mu a^\nu$$

For a boost with velocity v along the x^1 -axis, the matrix is:

$$\Lambda_\nu^\mu = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

where $\beta = v$ in natural units. Applying this to the four-momentum $p^\mu = (E, p_x, p_y, p_z)$ gives:

$$E' = \gamma(E - \beta p_x)$$

$$p'_x = \gamma(p_x - \beta E)$$

$$p'_y = p_y$$

$$p'_z = p_z$$